Seat No.: ________ Enrolment No.___________

GUJARAT TECHNOLOGICAL UNIVERSITY

BE –SEMESTER 1&2(NEW SYLLABUS)EXAMINATION- WINTER 2018

Subject Code: 3110005 **Date: 18-01-2019**

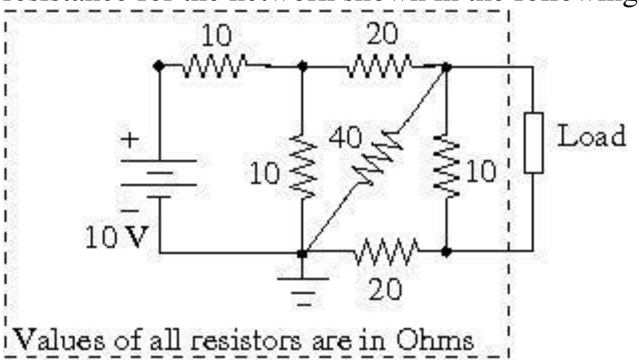
Subject Name: Basic Electrical Engineering

Time: 10:30 am to 01:00 pm **Total Marks: 70**

Instructions:

1. **Attempt all questions.**
2. **Make suitable assumptions wherever necessary.**
3. **Figures to the right indicate full marks.**

			Marks
Q.1	**(a)**	Define Amplitude, Frequency and Time period for alternating quantities.	**03**
	(b)	Briefly describe the operating principle of a transformer.	**04**
	(c)	Obtain the value of Norton's equivalent current and Norton's equivalent resistance for the network shown in the following figure.	**07**



Q.2	**(a)**	Prepare a list of parts of a DC machine. Explain any one part in detail.	**03**
	(b)	Briefly describe the auto transformer and its applications.	**04**
	(c)	The maximum values of voltage and current in a circuit are 400 V and 20 A respectively. Both the quantities are sinusoidal with 50 Hz frequency. The instantaneous values of voltage and current at time t=0 second are 283 V and 10 A respectively (both increasing and positive). Obtain the equations of voltage and current in this circuit at time 't' second. Also find out the active power consumption in the circuit.	**07**
		OR	
	(c)	In a series R-L circuit, a voltage of 10 V at 50 Hz frequency produces a current of 750 mA. In the same circuit with same magnitude of applied voltage with a frequency of 75 Hz produces a current of 500 mA. Find out the values of R and L in the circuit.	**07**
Q.3	**(a)**	Briefly describe pipe earthing.	**03**
	(b)	Mention the types of single phase induction motor. Explain any one of them.	**04**
	(c)	Derive the equations of active, reactive and apparent powers in a series R-L circuit with sinusoidal AC supply.	**07**
		OR	

Q.3	**(a)**	Give a list of safety devices used for home appliances.	**03**
	(b)	Give a comparison between squirrel cage induction motor and wound rotor induction motor.	**04**
	(c)	Derive the equations of capacitor voltage and circuit current in a series R-C circuit connected to a DC supply through a switch. Assume that switch is initially open and it is closed at time t=0 second.	**07**
Q.4	**(a)**	Discuss the difference between MCB and Fuse.	**03**
	(b)	Why the consumers should improve their power factor?	**04**
	(c)	Explain Thevenin's theorem. Take suitable example and explain the steps to apply Thevenin's theorem for a resistive circuit with a constant DC voltage source.	**07**
		OR	
Q.4	**(a)**	What is MCCB? Where is it used?	**03**
	(b)	Compute the monthly energy charges for an air conditioner having consumption of 2 kW. Daily usage of the air conditioner is 10 hours. Energy charges are Rs 8 per unit.	**04**
	(c)	Explain the term power factor. Explain the steps to obtain power factor of an AC circuit with parallel connection of R, L and C elements.	**07**
Q.5	**(a)**	Describe the stator construction of a single phase induction motor.	**03**
	(b)	Write a short note on Miniature Circuit Breaker (MCB)	**04**
	(c)	Explain the term rotating magnetic field with proper diagrams in case of a three phase induction motor.	**07**
		OR	
Q.5	**(a)**	Describe the construction of rotor for a slip ring type three phase induction motor.	**03**
	(b)	Write a short note on Earth Leakage Circuit Breaker (ELCB).	**04**
	(c)	Explain the working principles of a synchronous generator and a synchronous motor.	**07**

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Seat No.: ________ Enrolment No.___________

GUJARAT TECHNOLOGICAL UNIVERSITY

BE - SEMESTER–I &II (NEW) EXAMINATION – SUMMER-2019

Subject Code: 3110005 **Date: 20/06/2019**

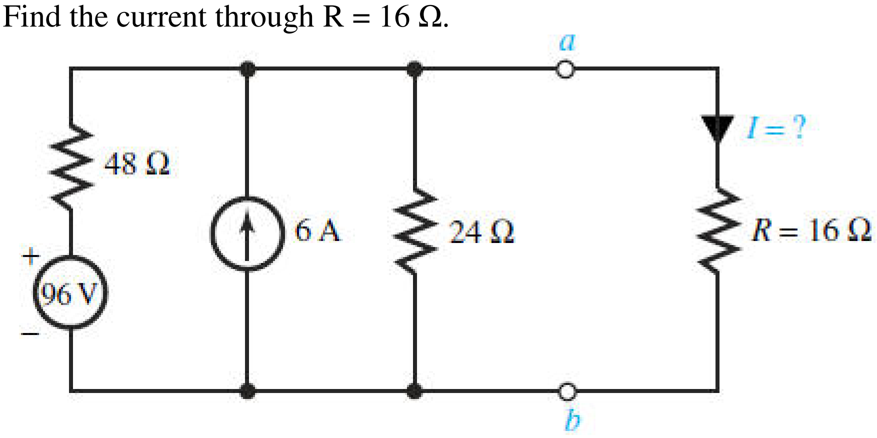
Subject Name: Basic Electrical Engineering

Time: 10:30 AM TO 01:00 PM **Total Marks: 70**

Instructions:

1. **Attempt all questions.**
2. **Make suitable assumptions wherever necessary.**
3. **Figures to the right indicate full marks.**

			Marks
Q.1	**(a)**	State and explain Kirchoff's voltage and current laws.	**03**
	(b)	Compare the resistive series and parallel circuit.	**04**
	(c)	Consider the circuit shown in Figure. Reduce the portion of the circuit to the left of terminals a–b to (a) a Thévenin equivalent and (b) a Norton equivalent. Find the current through R = 16 Ω.	**07**



Q.2	**(a)**	For series resonant circuit with brief description draw the phasor diagrams for following conditions (i) At resonant (ii) Below resonant (iii) Above resonant.	**03**
	(b)	Prove that the sum of readings of two watt meters connected to measure power in three phase circuit gives total power consumed by the circuit.	**04**
	(c)	A series RLC circuit with L= 160 mH, C= 100 µF and R = 40 Ω is connected to a sinusoidal voltage V(t) = 40 Sinωt, with ω= 200 rad/sec. Find (i) What is the Impedance of the circuit. (ii) Let the current at any instant in the circuit be I(t) = I_0 Sin(ωt-Φ). Find I_0 (iii) What is the Phase Φ ?	**07**
		OR	
	(c)	A balanced star connected load of (4+j3) Ω per phase is connected to a balance 3 phase 400 V supply. Find the line current, power factor, active power and reactive power.	**07**
Q.3	**(a)**	For A.C. sinusoidal current prove that $I_{rms} = 0.707\ I_m$.	**03**
	(b)	Explain voltage step-up and step-down operation in autotransformer with diagram.	**04**
	(c)	Explain various connections of three phase transformer with diagram.	**07**
		OR	
Q.3	**(a)**	Explain in brief single phase RC parallel circuit with phasor diagram	**03**
	(b)	Derive the E.M.F. equation of a single phase transformer.	**04**

	(c)	Explain with diagram construction of core type and shell type transformer.	**07**
Q.4	**(a)**	State significance of the back emf in DC motor.	**03**
	(b)	Classify and compare various DC motor.	**04**
	(c)	Explain construction of synchronous generator with diagram.	**07**
		OR	
Q.4	**(a)**	Give the classification of Induction motor.	**03**
	(b)	Discuss how the rotating magnetic field is produced in three phase induction motor.	**04**
	(c)	Explain the working of single phase induction motor with diagram.	**07**
Q.5	**(a)**	State function of various parts of HT cable.	**03**
	(b)	Give the comparison of fuse and MCB.	**04**
	(c)	Explain plate earthing with diagram.	**07**
		OR	
Q.5	**(a)**	What is power factor and why improvement is required in that?	**03**
	(b)	State and explain in brief important electrical characteristics of battery.	**04**
	(c)	Calculate the electricity bill amount for a month of April, if 4 bulbs of 40 W for 5 h, 4 tube lights of 60 W for 5 h, a TV of 100 W for 6 h, a washing machine of 400 W for 3 h, a water pump of 0.5 HP for 15 minutes are used per day. The cost per unit is Rs 3.50. Consider 1 HP = 746 watts	**07**

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Seat No.: ________ Enrolment No.___________

GUJARAT TECHNOLOGICAL UNIVERSITY

BE - SEMESTER– I & II (NEW) EXAMINATION – WINTER 2019

Subject Code: 3110005 **Date: 11/01/2020**

Subject Name: Basic Electrical Engineering

Time: 10:30 AM TO 01:00 PM **Total Marks: 70**

Instructions:

1. **Attempt all questions.**
2. **Make suitable assumptions wherever necessary.**
3. **Figures to the right indicate full marks.**

			Marks
Q.1	**(a)**	State Ohm's law and Kirchhoff's Laws in context with DC circuits.	**03**
	(b)	A 100V,60 Watt bulb is to be operated from a 220V supply. What is the resistance to be connected in series with the bulb to glow normally?	**04**
	(c)	Derive an expression for equivalent resistances of a star connected network to transform into a Delta connected network.	**07**
Q.2	**(a)**	State Thevenin's and Norton's Theorems.	**03**
	(b)	Compare series and parallel resonance in ac circuit.	**04**
	(c)	A single phase R-L-C circuit having resistance of 8Ω, inductance of 80mH and capacitance of 100μF is connected across single phase ac 150 V , 50Hz supply. Calculate the current, power factor and voltage drop across inductance and capacitance.	**07**
		OR	
	(c)	Derive an expression for the voltage across the capacitor during charging through the resistor at any instant $V_C = V(1-e^{-t/RC})$. Assume that RC series circuit is connected across a DC supply of voltage V.	**07**
Q.3	**(a)**	Define the following terms in connection with AC waveforms:- 1. Q-Factor 2. Power Factor 3.Form factor.	**03**
	(b)	Draw impedance triangle, Voltage triangle, Power triangle for single phase R-L series circuit.	**04**
	(c)	Derive the relationship between Phase and Line values of voltages and currents in case of 3-phase Star- connection.	**07**
		OR	
Q.3	**(a)**	Define the terms:- 1. Real power 2. Reactive power 3. Apparent power.	**03**
	(b)	Derive the EMF equation of single phase transformer.	**04**
	(c)	With neat circuit diagram and a phasor diagram prove that two watt meters are sufficient to measure total power in 3-phase system.	**07**
Q.4	**(a)**	Explain working principle of single phase Transformer.	**03**
	(b)	Give Merits, Demerits and Applications of Induction Motor.	**04**
	(c)	Explain in detail the construction of an Alternator.	**07**
		OR	
Q.4	**(a)**	Explain working principle of D.C. Motor.	**03**
	(b)	Compare poly phase Induction Motor and single phase Induction Motor.	**04**

	(c)	Explain Generation of Rotating Magnetic Field in 3-phase Induction Motor with diagrams and equations.	**07**
Q.5	**(a)**	Classify different types of cables with reference to voltage and insulation materials.	**03**
	(b)	Explain the terms:- 1. Residual magnetism 2. Coercive Force.	**04**
	(c)	Explain the process of charging and discharging of Lead acid cell.	**07**

OR

Q.5	**(a)**	Compare MCB and ELCB.	**03**
	(b)	Write safety precautions for electrical Applications.	**04**
	(c)	Explain different methods of power factor improvement.	**07**

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Seat No.: ________ Enrolment No.___________

GUJARAT TECHNOLOGICAL UNIVERSITY

BE – SEMESTER 1&2 EXAMINATION – SUMMER 2020

Subject Code: 3110005 **Date:10/11/2020**

Subject Name: Basic Electrical Engineering

Time: 10:30 AM TO 01:00 PM **Total Marks: 70**

Instructions:

1. **Attempt all questions.**
2. **Make suitable assumptions wherever necessary.**
3. **Figures to the right indicate full marks.**

			Marks
Q.1	**(a)**	A resistance of 10Ω is connected in series with two resistances each of the 15Ω arranged in parallel. What resistance must be shunted across this parallel combination so that total current taken shall be 1.5 A with 20 V applied?	**03**
	(b)	State the superposition theorem with suitable example.	**04**
	(c)	For the Wheatstone bridge diagram shown in **Figure 1**, obtain the current flowing through the 20Ω resistance using Thevenin's equivalent network.	**07**
Q.2	**(a)**	Explain Kirchoff's law for DC series network in brief.	**03**
	(b)	Define the following terms for AC (alternating current) signal: (i) Crest Factor (ii) Form Fator (iii) Average Value (iv) RMS Value.	**04**
	(c)	A current of 5 A flows through a non-inductive resistance in series with a choking coil when supplied at 250V,50Hz. If the voltage across the resistance is 125 V and across the coil 200V, calculate (i) impedance, reactance and resistance of the coil (ii) the power absorbed by the coil (iii) and total power.	**07**
		OR	
	(c)	Prove that the current in purely inductive circuit lags its voltage by 90° and average power consumption in pure inductor is zero.	**07**
Q.3	**(a)**	Write the comparison between series resonance and parallel resonance condition in AC circuit.	**03**
	(b)	Derive the relation between line-voltage and phase-voltage for three-phase four wire star connection network. Also, prove that the total three-phase power consumption in star connection is $P_T = \sqrt{3}\ V_L\ I_L \cos \varphi$.	**04**
	(c)	Explain various connections of three phase transformer with diagram.	**07**
		OR	
Q.3	**(a)**	Explain magnetic hysteresis.	**03**
	(b)	Derive the E.M.F. equation of a single phase transformer	**04**
	(c)	State the difference in core type and shell type transformer with neat and clean construction diagram.	**07**
Q.4	**(a)**	How the rotating magnetic field is produce in three-phase induction motor? Explain in brief.	**03**
	(b)	Why single-phase induction motor is not self starting while three-phase induction motor is self starting. Explain in brief.	**04**
	(c)	Explain construction of Alternator with neat diagram.	**07**
		OR	

Q.4	**(a)**	Justify that how back e.m.f. in DC motor acts like a governor.	**03**
	(b)	State the comparison of generator and motor action with respect to design and working principle. Draw the necessary diagram.	**04**
	(c)	Write working principle of DC motor with neat diagram.	**07**
Q.5	**(a)**	Calculate the resistance of a 100 m length of wire having a uniform cross sectional area of 0.02 mm^2 and having resistivity of 40 $\mu\Omega$-cm.	**03**
	(b)	Discuss types of cables used for residential and commercial wiring.	**04**
	(c)	Explain the following protective devices in detail: (i) SFU (ii) MCB (iii) ELCB	**07**
		OR	
Q.5	**(a)**	A d-c generator has an e.m.f of 200 volts and provides a current of 10 amps. How much energy does it provide each minute?	**03**
	(b)	Explain the construction of the lead-acid battery with neat diagram.	**04**
	(c)	Explain different types earthing and its importance in electrical utility system in detail.	**07**

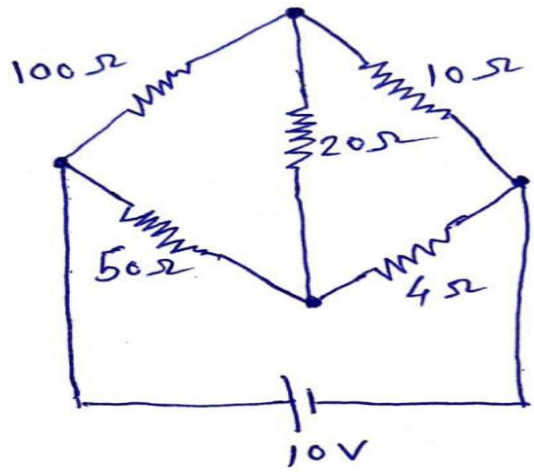

Figure - 1

Seat No.: ________ Enrolment No.___________

GUJARAT TECHNOLOGICAL UNIVERSITY

BE - SEMESTER–1/2 EXAMINATION – WINTER 2021

Subject Code:3110005 Date:29/03/2022

Subject Name:Basic Electrical Engineering

Time:10:30 AM TO 01:00 PM Total Marks:70

Instructions:

1. **Attempt all questions.**
2. **Make suitable assumptions wherever necessary.**
3. **Figures to the right indicate full marks.**
4. **Simple and non-programmable scientific calculators are allowed.**

Q.1	**(a)**	Calculate current, resistance and energy consumed by an oven rated 230V, 1KW when used for 20 hours. Also calculate the electricity bill at the rate of Rs. 7/- per unit.	03
	(b)	Define Power Factor. What are the disadvantages of Low Power Factor?	04
	(c)	Summarize; Mathematical & Waveform representation of an alternating sinusoidal quantity (voltage or current). Demonstrate time period, peak value, Peak to Peak value, Average value, RMS value. Write definition of all.	07
Q.2	**(a)**	Write statement of Superposition, Norton and Thevenin theorem.	03
	(b)	Write classification of Electric Networks.	04
	(c)	Write applications of Thevenin's theorem. Find the current passing through 2 Ω (Fig. A) resistance using Thevenin's theorem. All resistances are in Ω.	07
		OR	
	(c)	Derive the expression for the rise of current in R-L series circuit when a D.C. supply is switched on to it. Define time constant of it.	07
Q.3	**(a)**	Define Form factor. State the value of form factor for Sinusoidal waveform.	03
	(b)	A series resonance network consisting of a resistor of 30Ω, a capacitor of 2μF and an inductor of 20mH is connected across a sinusoidal supply voltage which has a constant output of 9 volts at all frequencies. Calculate, the resonant frequency, the current at resonance, the voltage across the inductor and capacitor at resonance, the Quality factor and the bandwidth of the circuit.	04
	(c)	With the help of waveforms and phasor diagrams, comment on phase relationship between voltage and current in single phase RLC series circuit.	07
		OR	
Q.3	**(a)**	Define Q-factor of RLC series circuit. What is the importance of it?	03
	(b)	Draw Impedance triangle and Admittance triangle.	04
	(c)	A balanced 3-phase load consists of 3 coils each of resistance of 6 Ω and inductive reactance of 8 Ω. Determine line current and power absorbed when the coils are 1. Star connected and 2. Delta connected across 400V, 3-phase supply.	07
Q.4	**(a)**	State advantages of polyphase systems.	03
	(b)	Mention Merits and Demerits of Induction Motor.	04

	(c)	Explain construction of single phase Transformer.	**07**

OR

Q.4	**(a)**	Write applications of Auto Transformer.	**03**
	(b)	Explain the principle of operation of an Alternator.	**04**
	(c)	Describe construction of a DC machine.	**07**
Q.5	**(a)**	Mention advantages of MCB over Fuse.	**03**
	(b)	Mention basic guidelines (Safety Rules) regarding safe handling of electricity.	**04**
	(c)	Explain construction of cable in detail with neat diagram. Label all the parts.	**07**

OR

Q.5	**(a)**	Briefly explain block diagram of ELCB.	**03**
	(b)	Mention important points regarding Earthing of Electrical Installation.	**04**
	(c)	Write short technical notes on Battery Maintenance. What precautions to be taken for Lead Acid Batteries?	**07**

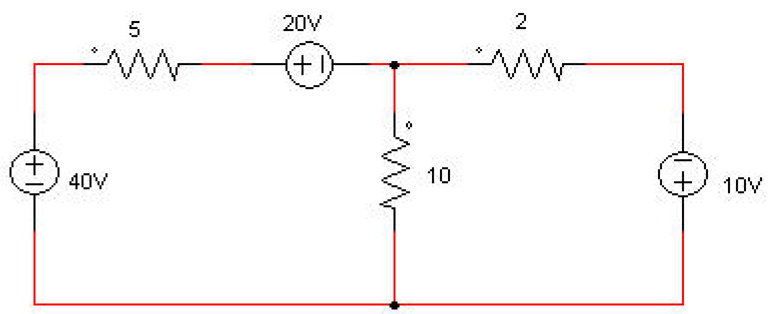

Fig. A

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Seat No.: ________ Enrolment No.___________

GUJARAT TECHNOLOGICAL UNIVERSITY

BE- SEMESTER–I & II(NEW)EXAMINATION – SUMMER 2022

Subject Code:3110005 Date:12-08-2022

Subject Name:Basic Electrical Engineering

Time:10:30 AM TO 01:00 PM Total Marks:70

Instructions:

1. **Attempt all questions.**
2. **Make suitable assumptions wherever necessary.**
3. **Figures to the right indicate full marks.**
4. **Simple and non-programmable scientific calculators are allowed.**

			Marks
Q.1	**(a)**	Calculate the Resistance of a 100 m length of wire having a uniform cross sectional area of 0.02 mm^2 and having resistivity of 40 $\mu\Omega$-cm.	**03**
	(b)	Explain Kirchoff's law for DC series network in brief.	**04**
	(c)	Define the following terms for AC (alternating current) signal: (i) Peak Factor (ii) Form Factor (iii) Average Value (iv) RMS Value (v) Time period (vi)Frequency (vii) Cycle	**07**
Q.2	**(a)**	State the Superposition theorem with suitable example.	**03**
	(b)	State the Norton's theorem with suitable example.	**04**
	(c)	For the Wheatstone bridge diagram shown in Figure 1, obtain the current flowing through the 20Ω resistance using Thevenin's equivalent network.	**07**
		OR	
	(c)	Derive an expression for the voltage across the capacitor during charging through the resistor at any instant $V_c = V(1-e^{-t/RC})$. Assume that RC series circuit is connected across a DC supply of voltage V.	**07**
Q.3	**(a)**	Write the comparison between Series resonance and Parallel resonance condition in AC circuit.	**03**
	(b)	Distinguish between (i) Apparent power (ii) Active power and (iii) Reactive power in ac circuits.	**04**
	(c)	Prove that the current in purely Capacitive circuit leads its voltage by 90° and average power consumption in pure capacitor is zero.	**07**
		OR	
Q.3	**(a)**	List out the merits of two-watt meter method.	**03**
	(b)	Draw Impedance triangle, Voltage triangle, Power triangle for single phase R-L series circuit.	**04**
	(c)	Obtain the relationship between line and phase values of current in a three phase, balanced, delta connected system.	**07**
Q.4	**(a)**	Explain working principle of single phase Transformer.	**03**
	(b)	Mention Merits and Demerits of Single Phase Induction Motor.	**04**
	(c)	Explain construction of Alternator with neat diagram.	**07**
		OR	
Q.4	**(a)**	Write applications of Auto Transformer.	**03**
	(b)	Compare Squirrel cage induction motor and Slip ring Induction Motor.	**04**
	(c)	Describe construction of a DC machine.	**07**
Q.5	**(a)**	Explain the protective device Switch Fuse unit in detail.	**03**
	(b)	With help of suitable labeled diagram, explain any two types of wires used in the residential and commercial wiring.	**04**

	(c)	Explain different methods of Power factor Improvement.	**07**

OR

Q.5	**(a)**	Compare MCB and ELCB.	**03**
	(b)	Write safety precautions for Electrical Appliances.	**04**
	(c)	Classify different types Earthing and explain any one in detail.	**07**

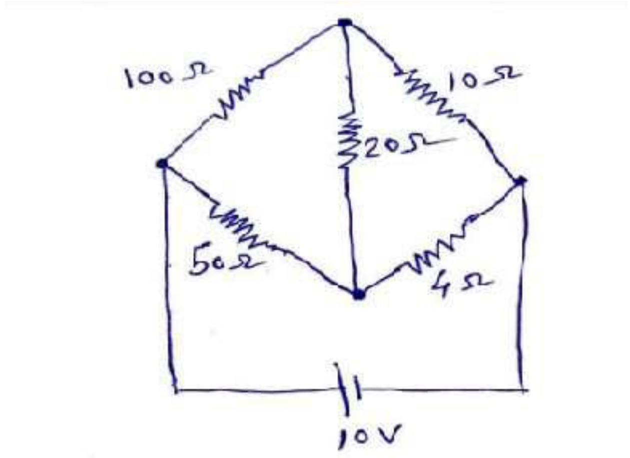

Figure - 1

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Seat No.: ________ Enrolment No.___________

GUJARAT TECHNOLOGICAL UNIVERSITY

BE - SEMESTER–I & II(NEW) EXAMINATION – WINTER 2022

Subject Code:3110005 Date:13-03-2023

Subject Name:Basic Electrical Engineering

Time:10:30 AM TO 01:00 PM Total Marks:70

Instructions:

1. **Attempt all questions.**
2. **Make suitable assumptions wherever necessary.**
3. **Figures to the right indicate full marks.**
4. **Simple and non-programmable scientific calculators are allowed.**

			Marks
Q.1	**(a)**	State and explain Kirchhoff's current Law	**03**
	(b)	Draw the construction of single phase core type transformer	**04**
	(c)	Derive an expression for equivalent resistances of a Delta connected network to transform into a Star connected network.	**07**
Q.2	**(a)**	Prepare a list of parts of a Single phase AC Motor.	**03**
	(b)	When three resistances of 10Ω, 20Ω, and 30Ω are connected in series across 230 V supply. Find (1) equivalent resistance (2) current flowing through each resistance, (3) voltage drop across each and (4) power loss in each resistor	**04**
	(c)	Describe phenomena of series resonance in R-L-C series circuit.	**07**
		OR	
	(c)	Prove that the average power consumption in a pure inductive circuit is zero.	**07**
Q.3	**(a)**	Write statement of Superposition, Thevenin's and Norton's Theorems.	**03**
	(b)	Derive an emf equation of single phase transformer	**04**
	(c)	Define following terms in connection with A.C wave forms : (i) Frequency (ii) Time period (iii) Instantaneous value (iv) Peak factor (v) form factor (vi) R. M. S. Value and (vii) Average Value	**07**
		OR	
Q.3	**(a)**	Write short note on autotransformer	**03**
	(b)	Compare series and parallel RLC circuit resonance	**04**
	(c)	Discuss how the rotating magnetic field is produced in three phase induction motor	**07**
Q.4	**(a)**	State types of sing phase split phase induction motor	**03**
	(b)	Explain Working of capacitor start Single phase induction motor	**04**
	(c)	List the various parts of DC Motor and explain each	**07**
		OR	
Q.4	**(a)**	Define with equation Active, reactive and apparent power in AC circuit	**03**
	(b)	What is power factor? State the methods of power factor improvement	**04**
	(c)	Prove that the sum of readings of two watt meters connected to measure power in three phase circuit gives total power consumed by the circuit	**07**
Q.5	**(a)**	Discuss the difference between MCB and Fuse	**03**
	(b)	Compute the monthly energy charges for an air conditioner having Power consumption of 3 kW and daily uses 10 hours. Energy charges Rs. 5 per unit	**04**
	(c)	Explain Working of Earth Leakage Circuit Breaker (ELCB).	**07**

OR

Q.5	**(a)**	Draw the structure of underground cable with name of all sections.	**03**
	(b)	Explain construction of 3-phasae induction motor with diagram	**04**
	(c)	Briefly describe pipe earthing	**07**

Seat No.: ________ Enrolment No.___________

GUJARAT TECHNOLOGICAL UNIVERSITY

BE- SEMESTER–I & II(NEW) EXAMINATION – SUMMER 2023

Subject Code:3110005 Date:08-08-2023

Subject Name:Basic Electrical Engineering

Time:10:30 AM TO 01:00 PM Total Marks:70

Instructions:

1. **Attempt all questions.**
2. **Make suitable assumptions wherever necessary.**
3. **Figures to the right indicate full marks.**
4. **Simple and non-programmable scientific calculators are allowed.**

			MARKS
Q.1	**(a)**	Determine the number of branches and nodes in the circuit shown in fig 1. Identify which elements are in series and which are in parallel.	**03**
	(b)	Find the average and rms value for the waveform shown in fig 2.	**04**
	(c)	Explain working of a single-phase transformer and derive the e.m.f. equation. Why the rating of transformer is in KVA?	**07**
Q.2	**(a)**	Can we apply KVL in a loop containing a current source? Give answer with reason.	**03**
	(b)	Explain ideal Voltage and current source	**04**
	(c)	Determine the current in the branches of the network in Fig.3 using nodal analysis.	**07**
		OR	
	(c)	A 120 V, 500 W lamp is used with a series choke on a 230 V, 60 Hz line, what is the inductance of the required choke if the choke has a Q of 2 ?	**07**
Q.3	**(a)**	Distinguish between a mesh and a loop of a circuit.	**03**
	(b)	What is the use of form factor and peak factor; also differentiate between EMF and potential difference.	**04**
	(c)	Determine the current through 2 ohm resistor connected between A and B in the circuit shown in fig.4 using Thevenin theorem	**07**
		OR	
Q.3	**(a)**	What is the need for power factor improvement in electrical circuits?	**03**
	(b)	Summarize the advantages of 3 phase circuits over single-phase circuits.	**04**
	(c)	A 400 v, 50 Hz 3-phase supply has 1 00 ohm between R and Y, j 1 00 ohm between Y and B and - j 100 ohm between B and R. Find (a) line currents for phase sequence RYB(b) star connected balanced resistors for the same power	**07**
Q.4	**(a)**	Draw and explain briefly BH curve.	**03**
	(b)	Write advantages of the sinusoidal waveform for electrical power applications.	**04**
	(c)	With the help of phasor diagram explain two wattmeter methods for the measurement of three-phase power.	**07**
		OR	
Q.4	**(a)**	What is the significance of back E.M.F. in a DC Motor?	**03**
	(b)	State advantages and disadvantages of autotransformer.	**04**
	(c)	What is the time constant of an RC circuit? Show that its units are the same as that of time. How this quantity signify about capacitor voltage during charging?	**07**
Q.5	**(a)**	Why single-phase induction motor is not self-starting?	**03**
	(b)	Analyze on how can the direction of rotation of a DC shunt motor be reversed	**04**
	(c)	Define the term battery capacity. On which factors it depends. Also list out the parts of lead acid battery	**07**
		OR	
Q.5	**(a)**	list out some safety measures against electric shocks	**03**
	(b)	What is the difference between wire & cable?	**04**

(c) Explain principle, construction and working of a synchronous generator in details. **07**

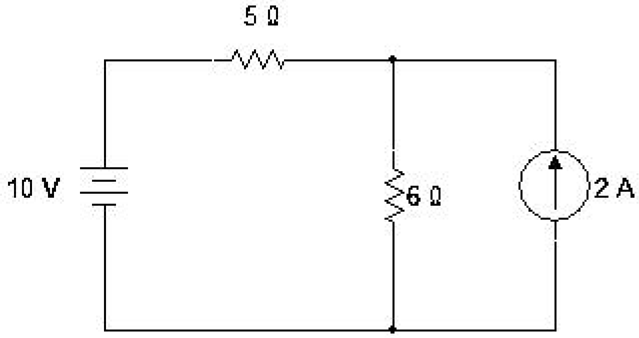


Fig1

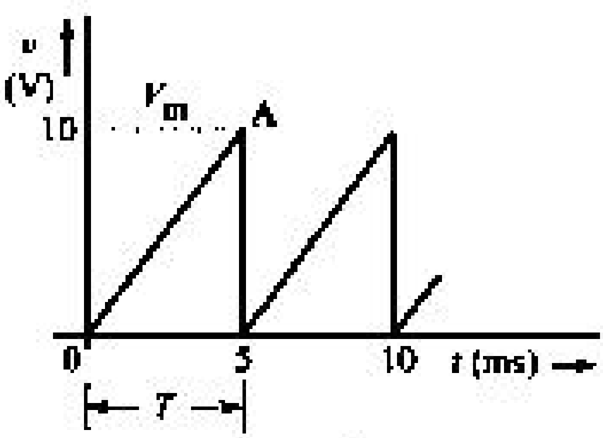


Fig2

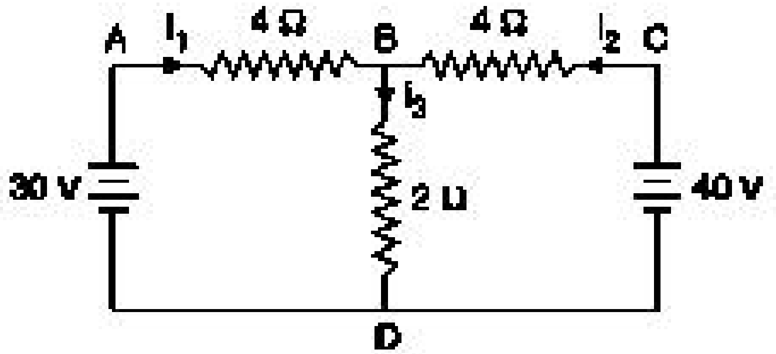


Fig.3

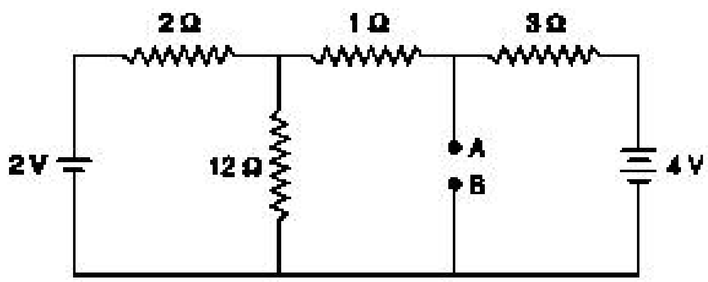


Fig.4

Seat No.: ________ Enrolment No.___________

GUJARAT TECHNOLOGICAL UNIVERSITY

BE - SEMESTER–I (NEW) EXAMINATION – WINTER 2023

Subject Code:3110005 Date:06-02-2024

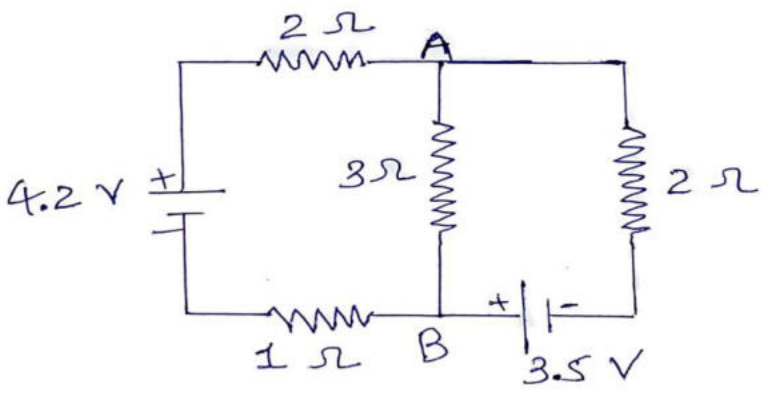
Subject Name:Basic Electrical Engineering

Time:02:30 PM TO 05:00 PM Total Marks:70

Instructions:

1. **Attempt all questions.**
2. **Make suitable assumptions wherever necessary.**
3. **Figures to the right indicate full marks.**
4. **Simple and non-programmable scientific calculators are allowed.**

			Marks
Q.1	**(a)**	Compare resistive series and parallel circuits.	**03**
	(b)	State the Thevenin's theorem with suitable example.	**04**
	(c)	Derive an expression for equivalent resistances of a star connected network to transform into a Delta connected network.	**07**
Q.2	**(a)**	Define Amplitude, Frequency and Time period for alternating quantities.	**03**
	(b)	Two coils are connected in parallel and a voltage of 200V is applied between the terminals. The total current taken by the circuit is 25 A and power dissipated in one of the coils is 1500 W. Calculate the resistance of each coil.	**04**
	(c)	Obtain the relationship between line and phase values of current in a three phase, balanced, delta connected system.	**07**
		OR	
	(c)	Prove that the current in purely inductive circuit lags its voltage by 90° and average power consumption in pure inductor is zero.	**07**
Q.3	**(a)**	Draw power triangle and define active power, reactive power and apparent power.	**03**
	(b)	If the waveform of a voltage has a form factor of 1.15 and peak factor of 1.5 and if the maximum value of a voltage is 4500 volts. Calculate the average and r.m.s. values of the voltage.	**04**
	(c)	Use the superposition theorem to calculate the current in branch AB of the circuit shown in below figure.	**07**



OR

Q.3	**(a)**	Explain working principle of synchronous motor.	**03**
	(b)	Classify and compare various types of D.C. motors.	**04**

	(c)	Explain construction of DC Machine.	**07**
Q.4	**(a)**	Derive the EMF equation of single-phase transformer.	**03**
	(b)	Compare core type and shell type single phase transformers.	**04**
	(c)	Explain various connections of three phase transformer with diagram.	**07**

OR

Q.4	**(a)**	Give a comparison between squirrel cage induction motor and wound rotor induction motor.	**03**
	(b)	Explain in brief working principle of Three Phase Induction Motor.	**04**
	(c)	Explain construction of synchronous generator with diagram.	**07**
Q.5	**(a)**	Compute the energy charges for an air conditioner having consumption of 2 kW for the month of April. Daily usage of the air conditioner is 12 hours. Energy charges are Rs 9 per unit.	**03**
	(b)	Write safety precautions for electrical Applications.	**04**
	(c)	Explain different methods of Power factor Improvement.	**07**

OR

Q.5	**(a)**	Write advantages and disadvantages of ELCB.	**03**
	(b)	Give comparison between MCB and Fuse.	**04**
	(c)	Classify different types of Earthing and explain Plate Earthing with diagram.	**07**

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Enrolment No./Seat No_______________

GUJARAT TECHNOLOGICAL UNIVERSITY

BE - SEMESTER–I & II (NEW) EXAMINATION – SUMMER 2024

Subject Code:3110005 **Date:03-07-2024**

Subject Name:Basic Electrical Engineering

Time:02:30 PM TO 05:00 PM **Total Marks:70**

Instructions:

1. **Attempt all questions.**
2. **Make suitable assumptions wherever necessary.**
3. **Figures to the right indicate full marks.**
4. **Simple and non-programmable scientific calculators are allowed.**

			Marks
Q.1	**(a)**	Define the following terms related to Electrical Circuits. 1. Branch 2. Node or Junction point 3. Mesh or Loop	**03**
	(b)	A 100 W,230 V lamp is connected in series with a 50W, 150 V lamp across 250 V supply mains. Calculate Voltage across each lamp.	**04**
	(c)	Explain Superposition theorem with suitable example.	**07**
Q.2	**(a)**	A R-L-C series circuit consists of R=10-ohm, L= 0.1 H and C= 50 μF. Find 1. Impedance 2. Current and 3. Active power, when it is connected to a a.c. source of 230V,50Hz.	**03**
	(b)	Draw the voltage and current waveforms of R-L and R-C series circuits.	**04**
	(c)	Derive the equation of three equivalent star connected resistances in terms of delta connected resistances.	**07**
		OR	
	(c)	Using Thevenin's theorem find current in branch BD of the network shown in Fig. 1	**07**
Q.3	**(a)**	Define the following terms in connection with A.C. wave forms. 1. Time period 2. R.M.S. Value 3. Average Value	**03**
	(b)	Three currents are represented by: $i_1 = 10 \sin \omega t$, $i_2 = 20 \sin (\omega t - \pi/6)$ and $i_3 = 30 \sin (\omega t + \pi/4)$. Find magnitude and phase angle of resultant current.	**04**
	(c)	Prove that current through pure inductor is always lagging by 90° to its voltage and power consumed is zero with necessary waveform and phasor diagram.	**07**
		OR	
Q.3	**(a)**	Write the characteristics of Series Resonant Circuit.	**03**
	(b)	Give advantages and disadvantages of Two Wattmeter method.	**04**
	(c)	Derive the relationship of voltages and currents for Star connection in 3-phase a.c. circuit.	**07**
Q.4	**(a)**	Derive E.M.F. equation of single-phase transformer.	**03**
	(b)	Compare core type and shell type transformers.	**04**
	(c)	Describe an auto transformer with its advantages, limitations and applications.	**07**
		OR	
Q.4	**(a)**	Give classification of D.C. Motors.	**03**
	(b)	Prepare list of different parts of D.C. machine and explain any one part with figure.	**04**
	(c)	Explain construction of synchronous generator with diagram.	**07**

Q.5	**(a)**	List the various safety devices used for domestic purpose.	**03**
	(b)	Explain necessity of earthing.	**04**
	(c)	Explain different methods for power factor improvement.	**07**

OR

Q.5	**(a)**	What is MCCB? Where it is used?	**03**
	(b)	State the advantages and applications of underground cables.	**04**
	(c)	Calculate the electricity bill amount for a month of March, if 5 bulbs of 40 W for 5 h, 6 tube lights of 60 W for 5 h, a TV of 100 W for 6 h, a washing machine of 400 W for 2 h, a water pump of 0.5 HP for 30 minutes are used per day. The cost per unit is Rs 3.50. Consider 1 HP = 746 watts	**07**

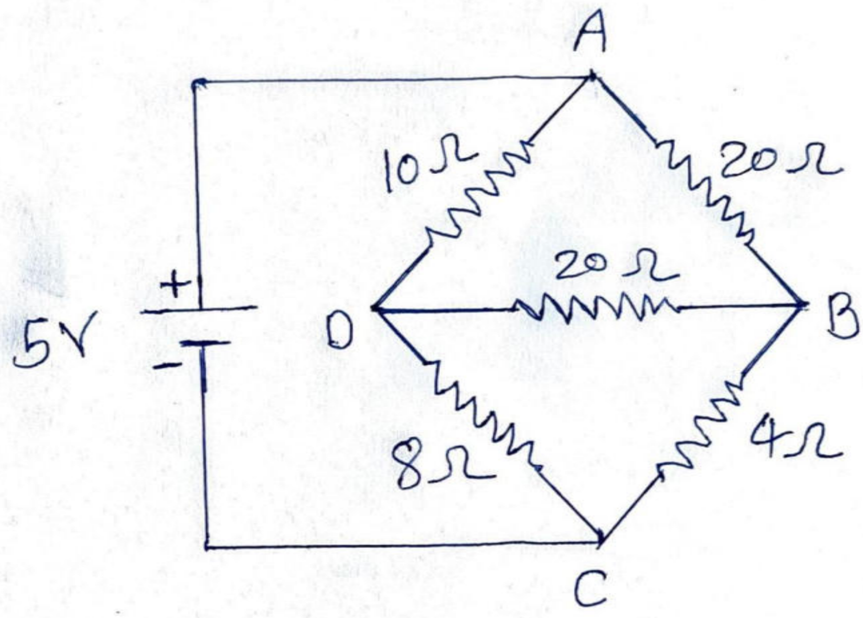

Fig: 1

Enrolment No./Seat No_______________

GUJARAT TECHNOLOGICAL UNIVERSITY

BE- SEMESTER–I & II EXAMINATION – WINTER 2024

Subject Code:3110005 **Date:13-01-2025**

Subject Name:Basic Electrical Engineering

Time:10:30 AM TO 01:00 PM **Total Marks:70**

Instructions:

1. **Attempt all questions.**
2. **Make suitable assumptions wherever necessary.**
3. **Figures to the right indicate full marks.**
4. **Simple and non-programmable scientific calculators are allowed.**

			MARKS
Q.1	**(a)**	State and explain Kirchhoff's current Law.	**03**
	(b)	Prepare a list of parts of a Single-phase AC Motor.	**04**
	(c)	State & Explain the Thevenin's theorem with suitable example.	**07**
Q.2	**(a)**	Write short note on autotransformer.	**03**
	(b)	Derive the EMF equation of single-phase transformer.	**04**
	(c)	When three resistances of 10Ω, 20Ω, and 30Ω are connected in series across 230 V supply. Find (1) equivalent resistance (2) current flowing through each resistance, (3) voltage drop across each and (4) power loss in each resistor	**07**
		OR	
	(c)	Derive the relationship between line and phase values of current in a three phase, balanced, delta connected system.	**07**
Q.3	**(a)**	Define following terms in connection with A.C wave forms: (i) Frequency (ii) Time period (iii) Amplitude.	**03**
	(b)	If the waveform of a voltage has a form factor of 1.11 and peak factor of 1.51 and if the maximum value of a voltage is 4500 volts. Calculate the average and r.m.s. values of the voltage.	**04**
	(c)	Describe the open circuit & short circuit test on single-phase transformer.	**07**
		OR	
Q.3	**(a)**	List out the merits of two-watt meter method.	**03**
	(b)	Compare series and parallel RLC circuit resonance.	**04**
	(c)	Derive an expression for the voltage across the capacitor during charging through the resistor at any instant V_c = V (1-e -t/RC). Assume that RC series circuit is connected across a DC supply of voltage V.	**07**
Q.4	**(a)**	Mention Merits and Demerits of Single-Phase Induction Motor.	**03**
	(b)	Compare Squirrel cage induction motor and Slip ring Induction Motor.	**04**
	(c)	Describe construction of a DC machine.	**07**
		OR	
Q.4	**(a)**	Write applications of Auto Transformer.	**03**
	(b)	Explain Working of capacitor start Single phase induction motor	**04**
	(c)	Explain construction of synchronous generator with diagram.	**07**
Q.5	**(a)**	Write advantages and disadvantages of ELCB.	**03**
	(b)	Draw the structure of underground cable with name of all sections.	**04**
	(c)	Explain different methods of Power factor Improvement	**07**

OR

Q.5	**(a)**	Write safety precautions for electrical Applications	**03**
	(b)	Compute the monthly energy charges for an air conditioner having Power consumption of 3 kW and daily uses 10 hours. Energy charges Rs. 5 per unit	**04**
	(c)	Classify different types Earthing and explain any one in detail.	**07**

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Enrolment No./Seat No_______________

GUJARAT TECHNOLOGICAL UNIVERSITY

BE- SEMESTER–I&II EXAMINATION – SUMMER 2025

Subject Code:3110005 **Date:01-07-2025**

Subject Name:Basic Electrical Engineering

Time:10:30 AM TO 01:00 PM **Total Marks:70**

Instructions:

1. **Attempt all questions.**
2. **Make suitable assumptions wherever necessary.**
3. **Figures to the right indicate full marks.**
4. **Simple and non-programmable scientific calculators are allowed.**

Q.1	**(a)**	State Ohm's law and Kirchhoff's laws.	**03**
	(b)	Define power factor. Write the disadvantages of low power factor.	**04**
	(c)	Derive the equation of star to delta and delta to star transformation.	**07**
Q.2	**(a)**	Write the advantages and disadvantages of polyphase systems.	**03**
	(b)	Define: (i) Form factor (ii) Peak factor (iii) Frequency and (iv) Cycle.	**04**
	(c)	State and explain Thevenin's theorem with suitable example.	**07**
		OR	
	(c)	A 100 V, 100-Watt lamp is connected in series with a 100 V, 60-watt lamp across 200 V supply. Determine current drawn and power consumed by each lamp.	**07**
Q.3	**(a)**	Define: (i) Active power (ii) Reactive power and (iii) apparent power in ac circuits.	**03**
	(b)	Explain the comparison between series resonance and parallel resonance condition in ac circuits.	**04**
	(c)	Prove that current in purely inductive circuit lags its voltage by 90 degree and average power consumption in pure inductive circuit is zero.	**07**
		OR	
Q.3	**(a)**	Define and explain RMS value and average value in ac circuits.	**03**
	(b)	Derive an expression for impedance, current and power factor for an R-L-C series circuit when supplied with ac voltage. Also draw the phasor diagram.	**04**
	(c)	Derive the relationship between line voltage and phase voltage, line current and phase current in a 3-phase balanced star connected circuit	**07**
Q.4	**(a)**	Define transformer. Write the main parts of transformer.	**03**
	(b)	Compare auto transformer with two winding transformers.	**04**
	(c)	Explain various connections of 3-phase transformers with necessary diagrams.	**07**
		OR	
Q.4	**(a)**	Write the advantages and disadvantages of auto transformer.	**03**
	(b)	Write the types of 1-phase induction motors. Explain any one of them.	**04**
	(c)	Explain construction and working of synchronous generator with necessary diagram.	**07**
Q.5	**(a)**	Compare squirrel cage induction motor with slip ring induction motor.	**03**
	(b)	Explain the types of dc motors with diagrams.	**04**
	(c)	Define earthing. Explain plate earthing with diagram.	**07**
		OR	
Q.5	**(a)**	Give a list of safety devices used for home appliances.	**03**
	(b)	Write the comparison of Fuse, MCB and ELCB.	**04**
	(c)	A series R-L-C circuit having a resistance of 8 Ω, inductance of 80 mH and capacitance of 100 μF is connected across a 150 Volt, 50 Hz supply. Calculate (i) the current (ii) the power factor and (iii) the voltage drops in the coil and capacitance.	**07**

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Enrolment No./Seat No_______________

GUJARAT TECHNOLOGICAL UNIVERSITY

BE- SEMESTER–I & II EXAMINATION – WINTER 2025

Subject Code:3110005 **Date:12-01-2026**

Subject Name:Basic Electrical Engineering

Time:02:30 PM TO 05:00 PM **Total Marks:70**

Instructions:

1. **Attempt all questions.**
2. **Make suitable assumptions wherever necessary.**
3. **Figures to the right indicate full marks.**
4. **Simple and non-programmable scientific calculators are allowed.**

Q.1	**(a)**	Calculate current, resistance and energy consumed by an electric iron rated 230 V, 2 KW when used for 12 hours. What is the monthly energy bill at Rs. 5/- per unit?	**03**
	(b)	Draw various types of alternating waveforms. Label peak, peak-to-peak, RMS and average value on sinusoidal AC waveform. Why are sinusoidal voltages preferred over other alternating voltages?	**04**
	(c)	Write basic guidelines (Safety Rules) regarding the safe handling of electricity and electrical appliances. Mention various types of electrical safety devices.	**07**
Q.2	**(a)**	A 100 V,100 W lamp is connected in series with a 100 V, 60W lamp across 200 V supply. Determine current drawn and power consumed by each lamp.	**03**
	(b)	What are the limitations of the superposition theorem?	**04**
	(c)	Find current in 2 Ω resistance by Thevenin's theorem circuit shown in figure 1. Also draw Thevenin's equivalent circuit.	**07**
		OR	
	(c)	Derive the expression for the voltage across the capacitor at any instant after the application of DC voltage V to a circuit with a capacitance C in series with resistance R. Define the time constant (τ) of the series RC circuit.	**07**
Q.3	**(a)**	A balanced star connected load of 4+j3 Ω per phase is connected to a balanced 3-phase 400 V supply. Find the active power and reactive power.	**03**
	(b)	Compare single phase alternating R-L circuit with R-C circuit.	**04**
	(c)	Write short technical note on series RLC resonance. Plot the variation of impedance, current and power factor with respect to the signal frequency.	**07**
		OR	
Q.3	**(a)**	A single-phase series RLC circuit having 50 Ω resistance, 0.1 H inductance and 50μF capacitance and it is connected with single phase 230 V, 50 Hz supply. Calculate the active power consumed by the circuit.	**03**
	(b)	Draw Power triangle and impedance triangle for single phase R-L series circuit and write the formula of power factor from it.	**04**
	(c)	Define power factor. What are the drawbacks of low power factor? Explain any one method to improve the power factor.	**07**

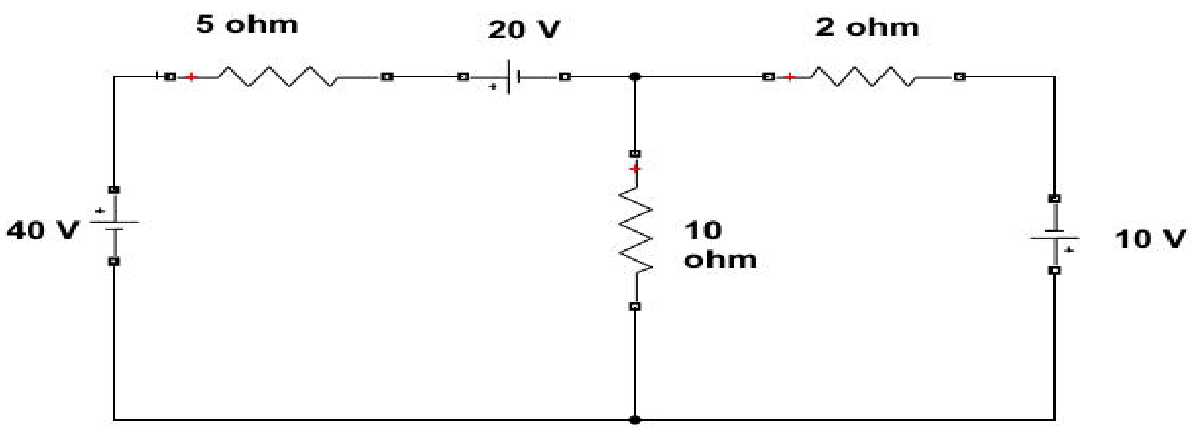
Q.4	**(a)**	Compare core type and shell type transformer.	**03**
	(b)	Write short note on B-H curve of magnetic materials.	**04**
	(c)	Explain the principle of operation of the transformer. Draw the construction of transformer and write function of each part.	**07**

OR

Q.4	**(a)**	Explain the principle of operation of induction motor.	**03**
	(b)	Mention the advantages of induction motor over dc motor.	**04**
	(c)	Draw construction of DC machine, label each part and write the function of each.	**07**
Q.5	**(a)**	Explain the construction of the cable.	**03**
	(b)	What is the necessity of earthing?	**04**
	(c)	Differentiate fuse, MCB and MCCB.	**07**

OR

Q.5	**(a)**	Compare equipment earthing with neutral earthing.	**03**
	(b)	Write a short technical note on battery maintenance.	**04**
	(c)	Elaborate various methods of earthing.	**07**



(Q. 2-C) Fig. 1